

Moving plates

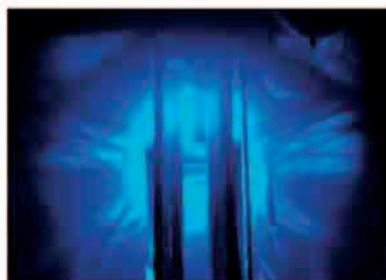
Earth's crust is always on the move, driven by heat currents that rise from deep within the planet. The movement is extremely slow, but hugely powerful. It has ripped the brittle lithosphere into great rocky plates that are pulling apart in some places and grinding together in others. These movements trigger earthquakes and volcanoes, especially in the frontier zones where the plates meet.

MOBILE MANTLE

The heat generated deep within the planet rises through the mantle to the surface, just like heat rising through soup simmering on a kitchen stove. It drives convection currents that circulate through the solid but very hot mantle rock, and keep it moving very slowly. The currents make the mantle rock rise to beneath the crust, spread sideways, then sink again as it cools.

HEAT SOURCE

Deep in Earth's mantle, uranium and other radioactive elements release nuclear energy in the form of heat. The process is the same as the reaction that takes place in this nuclear reactor. Some elements, such as uranium-238, have been generating heat in this way since the planet was formed.



▲ NUCLEAR ENERGY

Uranium fuel rods in a nuclear reactor radiate the same type of energy as the uranium inside the planet.

